

scikit-covtest: Covariance Matrix Hypothesis Testing in Python

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Abstract

Covariance matrices play a central role across diverse domains such as finance, machine learning, neuroscience, and genetics, where they are used for tasks including dimensionality reduction, connectivity inference, and risk estimation. Many of these applications require testing whether a covariance matrix follows a specific structure. While several R packages provide partial coverage of such tests, Python, an important language in machine learning, lacks a comprehensive, well-tested implementation. To address this gap, we introduce **scikit-covtest**, a Python package implementing a variety of hypothesis tests for covariance matrices spanning four categories: identity, sphericity, proportionality, and two-sample equality. The package provides a consistent **scipy**-style API, extensive documentation, and supporting functionality for multiple testing correction, synthetic data generation, and diagnostic evaluation. **scikit-covtest** is open source, available through PyPI, and lowers the barrier to applying modern covariance-testing methods in scientific applications.

Keywords: covariance matrix, hypothesis testing, high-dimensional statistics, multiple hypothesis testing, Python.

1. Introduction

Covariance matrices play a central role in modern statistics, with applications spanning finance (Pafka and Kondor 2003), machine learning (Minh and Murino 2017), neuroscience (Varoquaux, Gramfort, Poline, and Thirion 2010), and genetics (Head, Leslie, Cutler, and Epstein 2023; Hwang, Talbot, Head, Trevino, Wingo, and Kotlar 2026). They are used for dimensionality reduction (Bishop 2006), network discovery (Zhong, Wang, Lu, Zhang, Jiao, and Liu 2009), and quantifying risk (Ledoit and Wolf 2022). In genetics, covariance matrices are critical in characterizing the relationships between genetic variants (Xu 2013). As one of the fundamental quantities in the field, they are used in a variety of tasks such as gene-trait associations (Li, Liu, Bakshi, Jiang, Chen, Zheng, Sullivan, Visscher, Wray, Yang, and Zang 2023), polygenic risk score estimation (Ge, Chen, Ni, Feng, and Smoller 2019), and fine mapping (Wang, Sarkar, Carbonetto, and Stephens 2020). Hypothesis testing of covariance matrix structure is therefore a fundamental task in these domains, both in the classical large sample size domain and the high-dimensional domain where the number of variables is comparable or exceeds sample size.

Despite the importance of testing, and the extensive work developing methods for doing so, software implementations of these methods are fragmented and incomplete. In R, several

packages provide implementations of specific procedures (Ding and Hu 2025; Cao, He, and Zhou 2018; Barnard and Young 2018). However, these packages omit many methods, maintenance levels vary, and some have unresolved implementation issues documented in their repositories. In Python, functionality is even more limited: **statsmodels** (Seabold, Perktold *et al.* 2010) includes a small number of likelihood-based procedures and **Pingouin** (Vallat 2018) implements the classic Box’s M test (Box 1949), but to our knowledge, there is no unified, well-tested, and well-documented framework dedicated to covariance matrix hypothesis testing.

We introduce **scikit-covtest**, a Python package that provides a comprehensive, consistent, and tested implementation of a broad class of tests. These tests are organized into four main categories covering most common testing scenarios: (i) one-sample tests of a specified covariance matrix, (ii) sphericity tests, (iii) proportionality tests between two covariance matrices, and (iv) two-sample tests for equality of covariance matrices. The design follows a **SciPy**-style API, with standardized input conventions, clear return types, extensive documentation, and rigorous testing.

Beyond hypothesis tests, **scikit-covtest** includes supporting functionality essential in many applications. Multiple testing corrections (Benjamini and Hochberg 1995; Storey and Tibshirani 2003; Benjamini and Yekutieli 2001) are critical when many tests are conducted simultaneously, a common occurrence in the era of genome-wide studies. Synthetic data generation with controlled covariance structures enables reproducible benchmarking and method validation. Diagnostic functions for assessing multivariate normality (Mardia 1970) are also included, as normality is a common and critical assumption in many implemented tests. The package also includes curated datasets (LeCun, Bottou, Bengio, and Haffner 1998; Fiorini 2016), benchmarking utilities to evaluate test properties, and visualization tools to facilitate empirical evaluation and interpretation of the results.

Due to the broad usage of covariance matrices, different tests have been designed for specific regimes and alternatives. The most important distinctions are (i) whether the test is either classical or high-dimensional and (ii) whether the test is tuned for sparse or dense alternatives. Tests designed for one regime may be poorly calibrated, undefined, or low-power in another. Furthermore, many classical tests assume multivariate normality but differ in sensitivity to violations of this assumption. To illustrate these differences, we provide the outputs of the tests under standardized simulations to assist in determining which tests are appropriate for a given application.

The contents of this paper are as follows. In Section 2 we provide the mathematical background on hypothesis testing and describe the four classes of tests in this work. Section 3 describes the methods currently implemented in **scikit-covtest**, including the supporting methods for data generation, evaluation, and multiple hypothesis testing mentioned above. Section 4 evaluates null calibration and empirical power across all four hypothesis classes. Section 5 demonstrates usage on two of the bundled datasets. Finally, Section 6 provides some suggestions on future improvements and work. This package is installable via pip and is publicly available at <https://github.com/bystrogenomics/scikit-covtest>, along with the code to reproduce all figures and analyses in this work.

2. Background: Hypothesis Testing of Covariance Matrices

2.1. Notation and Problem Formulation

Vectors are denoted by bold lowercase letters (e.g., $\mathbf{x}$) and matrices by bold uppercase letters (e.g., $\mathbf{X}$). For a matrix $\mathbf{A} = (a_{ij})$, its transpose is $\mathbf{A}^\top$. The covariance matrix is denoted by $\mathbf{\Sigma} = (\sigma_{ij})$.

In the one-sample setting, let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be i.i.d. p -dimensional random vectors with mean $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}_k]$ and covariance matrix $\mathbf{\Sigma} = \text{Cov}(\mathbf{x}_k)$. The sample mean and covariance matrix are

$$\bar{\mathbf{x}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k, \quad \hat{\mathbf{\Sigma}} = \frac{1}{n} \sum_{k=1}^n (\mathbf{x}_k - \bar{\mathbf{x}})(\mathbf{x}_k - \bar{\mathbf{x}})^\top. \quad (1)$$

In the two-sample setting, we observe independent samples $\{\mathbf{x}_1, \dots, \mathbf{x}_{n_1}\}$ and $\{\mathbf{y}_1, \dots, \mathbf{y}_{n_2}\}$ with covariance matrices $\mathbf{\Sigma}_1$ and $\mathbf{\Sigma}_2$, and corresponding estimators $\hat{\mathbf{\Sigma}}_1$ and $\hat{\mathbf{\Sigma}}_2$.

We consider the following hypotheses for testing covariance structures:

- **Identity test:** Tests whether the population covariance matrix equals the identity,

$$H_0 : \mathbf{\Sigma} = \mathbf{I}_p, \quad (2)$$

which corresponds to variables being uncorrelated with unit variances; note that this constrains only the second moments and does not restrict the mean. This generalizes to $H_0 : \mathbf{\Sigma} = \mathbf{\Sigma}_0$ for known $\mathbf{\Sigma}_0$ either by replacing $\mathbf{I}_p$ with $\mathbf{\Sigma}_0$ or, equivalently, by whitening the data through $\mathbf{\Sigma}_0^{-1/2} \hat{\mathbf{\Sigma}} \mathbf{\Sigma}_0^{-1/2}$ and testing against $\mathbf{I}_p$, with calibration following the transformed statistic. Applications include monitoring for changes in machine alignment (Li, Wang, and Yeh 2013) and testing for intrasubject variation in high-dimensional gene expression data (Qayed and Han 2021).

- **Sphericity test:** Tests whether the covariance matrix is proportional to the identity,

$$H_0 : \mathbf{\Sigma} = c \mathbf{I}_p, \quad c > 0, \quad (3)$$

implying that all variables have equal variance and are mutually uncorrelated. Sphericity tests arise in repeated-measures ANOVA, where $c \mathbf{I}_p$ is a sufficient condition for valid within-subject comparisons (the exact requirement being sphericity of the within-subject contrasts) (Winer, Brown, Michels *et al.* 1971; Smith 2017). They also appear in signal processing applications such as determining the number of signal sources in radar systems (Liu, Sun, and Zhao 2017; Yuan and Zhang 2023).

- **Proportionality test:** Tests whether two covariance matrices differ only by a scalar multiple,

$$H_0 : \mathbf{\Sigma}_1 = c \mathbf{\Sigma}_2, \quad c > 0, \quad (4)$$

indicating the same correlation structure but potentially different overall scale. This has been used to test genetic variance matrices as an intermediate hypothesis between complete equality and completely unrelated structures (Phillips and Arnold 1999) and to test specific evolutionary patterns when overall variance is not of interest (Ackermann 2002).

- **Two-sample equality test:** Tests whether two populations share the same covariance structure,

$$H_0 : \Sigma_1 = \Sigma_2, \quad (5)$$

corresponding to equal variance and correlation patterns across samples. Applications include detecting genotype-phenotype interactions (Wang, Zhang, Zeng, Wu, Kemper, Xue, Zhang, Powell, Goddard, Wray *et al.* 2019), identifying parent-of-origin effects (Head *et al.* 2023), and detecting changes in financial portfolio variances (Li, Li, and Wang 2025).

2.2. Common Test Statistics

Many modern hypothesis tests are based on norms to measure distance between the sample covariance(s) and the hypothesized structure. For example, a single-sample test would be

$$T = \|\hat{\Sigma} - \mathbf{I}_p\|^2. \quad (6)$$

The choice of norm determines the alternatives that the test is sensitive to. For example, the Frobenius norm is sensitive to dense alternatives where a large proportion of entries differ from the null. This behavior can be seen when this norm is written in its alternate form as

$$\|\hat{\Sigma} - \mathbf{I}_p\|_F^2 = \sum_{i=1}^p (\hat{\sigma}_{ii} - 1)^2 + 2 \sum_{1 \leq i < j \leq p} \hat{\sigma}_{ij}^2. \quad (7)$$

Alternatively, the operator norm,

$$\|\hat{\Sigma} - \mathbf{I}_p\|_{\text{op}} = \max_{\|\mathbf{u}\|_2=1} |\mathbf{u}^\top (\hat{\Sigma} - \mathbf{I}_p) \mathbf{u}|, \quad (8)$$

will target departures in a single direction and is sensitive to low-rank structure. Finally, the entrywise maximum norm $\|\mathbf{A}\|_{\max} = \max_{1 \leq i, j \leq p} |a_{ij}|$ (the L_∞ norm of the vectorized matrix) targets sparse deviations. Two-sample tests function similarly, except the norm is taken between the two empirical covariance matrices.

Sphericity tests require modification to account for the unknown scale c . Centering the sample covariance about its best scalar approximant removes the identity component but not the scale itself, so the statistic must additionally be normalized by an estimate of c to obtain a null distribution free of the nuisance scale. Define the centered sample covariance matrix

$$\tilde{\Sigma} = \hat{\Sigma} - \frac{\text{tr}(\hat{\Sigma})}{p} \mathbf{I}_p, \quad \hat{c} = \frac{\text{tr}(\hat{\Sigma})}{p}. \quad (9)$$

A scale-invariant Frobenius-type sphericity statistic is

$$T_F^{\text{sph}} = \frac{\|\tilde{\Sigma}\|_F^2}{p \hat{c}^2} = \frac{p \text{tr}(\tilde{\Sigma}^2)}{\text{tr}(\hat{\Sigma})^2} - 1, \quad (10)$$

which coincides with the classical statistics of John (1971) and Ledoit and Wolf (2002) and whose null distribution does not depend on c . Max-type sphericity tests target sparse alternatives by examining the largest standardized entry of $\tilde{\Sigma}$,

$$T_{\max}^{\text{sph}} = \max_{1 \leq i \leq j \leq p} \frac{|\tilde{\sigma}_{ij}|}{\widehat{\text{se}}(\tilde{\sigma}_{ij})}, \quad (11)$$

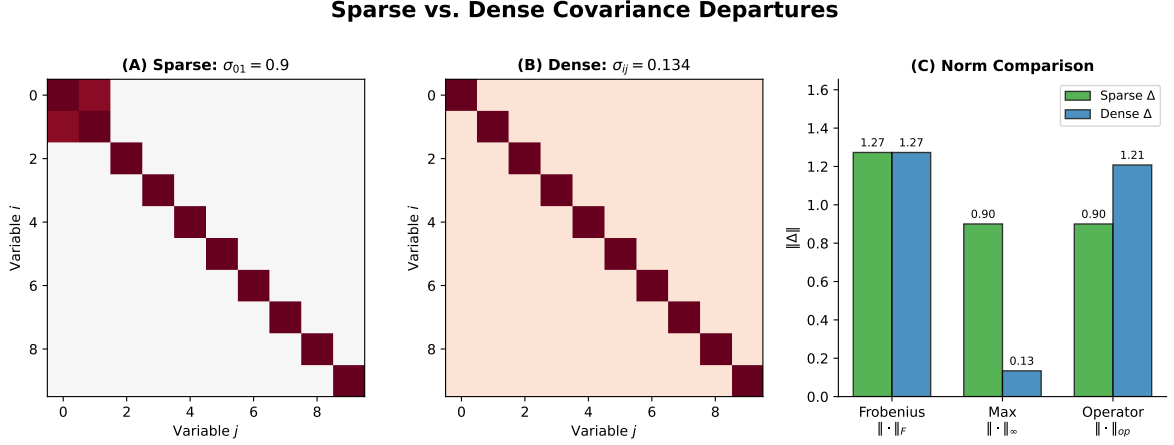


Figure 1: Illustration of sparse versus dense alternatives to the identity hypothesis, shown for $p = 10$ with departure matrix $\Delta = \Sigma - \mathbf{I}_p$. (A) Sparse alternative with a single large off-diagonal correlation ($\sigma_{12} = 0.9$). (B) Dense alternative with uniform small correlations ($\sigma_{ij} = 0.134$, $i \neq j$), calibrated to have the same Frobenius norm as (A). (C) Comparison of matrix norms. The two departure matrices have equal Frobenius norm by construction, but the entrywise maximum norm is $6.7\times$ larger for the sparse case, while the operator norm is $1.3\times$ larger for the dense case. This illustrates why entrywise-maximum (L_∞) based tests are preferred for detecting sparse departures, while Frobenius-based tests are sensitive to dense alternatives and operator-norm tests focus on low-rank structure.

where each entry is divided by an estimate of its null standard error. This standardization is necessary because, under sphericity, diagonal entries of the sample covariance have approximately twice the variance of the off-diagonal entries, so an unstandardized maximum would combine non-comparable coordinates; the standardized maximum is scale-invariant and is calibrated using an extreme-value (Gumbel) limit (Cai, Liu, and Xia 2013). Figure 1 illustrates the distinction between norms: two alternatives with identical Frobenius norms can have substantially different maximum-entry and operator norms.

In the classical regime where n is large relative to p , traditional distributional convergence, either to a chi-squared (Ledoit and Wolf 2002; Hallin and Paindaveine 2006; Muirhead 1982) or normal (Ahmad and von Rosen 2015; Li *et al.* 2025) distribution, can be used. However, when the data are high dimensional, tests often require bias correction on the test statistic to obtain proper size control and power (Ledoit and Wolf 2002; Srivastava 2007).

Beyond distributional assumptions, test robustness to heavy tails is often desirable. Robust variants replace $\hat{\Sigma}$ by heavy-tail-resistant scatter matrices, including procedures inspired by robust T^2 (O’Brien 1992) and tests based on Tyler’s M -estimator (Li *et al.* 2025). Because Tyler’s estimator identifies the elliptical shape only up to a positive scalar, such procedures are naturally suited to normalized shape, sphericity, and proportionality hypotheses; testing an absolute-scale null additionally requires a separate robust scale estimate.

2.3. Hypothesis Testing Diagnostics

To evaluate test calibration, we implement diagnostics for p-value uniformity under the null hypothesis. Classical goodness-of-fit tests assess whether p-values behave as expected under the null. The Kolmogorov-Smirnov test compares the empirical distribution to $\text{Uniform}(0, 1)$, while an Anderson-Darling statistic (Lehmann and Romano 2005) targets deviations in the distribution tails. Scalar measures quantify global discrepancies between the empirical and theoretical cumulative distribution functions. We note that exact uniformity is expected only for continuous, exactly calibrated p-values; discrete, permutation, and Monte Carlo p-values are typically only super-uniform, and the standard reference distributions for these goodness-of-fit statistics assume independence across the tested hypotheses.

Large-scale testing applications require additional diagnostics. The genomic control inflation factor λ_{GC} (Devlin and Roeder 1999) detects systematic test statistic inflation via the median χ^2 statistic. Storey’s π_0 estimator (Storey and Tibshirani 2003) infers the proportion of true nulls by examining the tail behavior of p-values. Both are descriptive summaries whose interpretation depends on signal prevalence: inflation or a low estimated π_0 can reflect widespread true signal rather than miscalibration alone. Together, these methods provide both formal tests and descriptive summaries for evaluating test behavior in high-dimensional settings.

2.4. Multiple Hypothesis Testing

Multiple hypothesis testing is essential in high-dimensional data analysis, particularly in omics fields where researchers routinely evaluate thousands to millions of hypotheses (Dudoit and Van Der Laan 2008). In genome-wide association studies (GWAS), for example, millions of SNPs are tested simultaneously, and without appropriate corrections, false positives would overwhelm true findings (Visscher, Brown, McCarthy, and Yang 2012).

Standard correction methods include the false discovery rate (FDR) (Benjamini and Hochberg 1995) and family-wise error rate (FWER) controls such as the Bonferroni adjustment. The widely used GWAS threshold of $p < 5 \times 10^{-8}$ exemplifies how multiple-testing corrections have been scaled to genomic contexts (Pe’er, Yelensky, Altshuler, and Daly 2008). These procedures help ensure that high-throughput studies yield reproducible, reliable results, minimizing wasted resources on spurious associations (Dudbridge and Gusnanto 2008; Storey and Tibshirani 2003).

To support both classical and modern testing contexts, the package implements several FDR control strategies, each valid under specific dependence conditions. The Benjamini-Hochberg (BH) (Benjamini and Hochberg 1995) procedure controls the FDR under independence and under positive regression dependence on a subset (PRDS), while the Benjamini-Yekutieli (BY) (Benjamini and Yekutieli 2001) procedure guarantees control under arbitrary dependence at the cost of a logarithmic penalty factor. Extensions such as weighted BH (Genovese, Roeder, and Wasserman 2006) incorporate prior information through nonnegative weights fixed independently of the p-values to improve power in heterogeneous settings. Storey-Tibshirani’s adaptive method (Storey and Tibshirani 2003) further increases power by estimating the null proportion π_0 , under corresponding assumptions on the estimator and dependence structure.

2.5. Testing Distributional Assumptions

Many classical covariance tests assume Gaussian or elliptical distributions and well-conditioned covariance matrices. To evaluate these assumptions, we implement several diagnostic methods. Eigenvalue spectrum analysis detects deviations from white noise or low-rank structure. The package computes the sample covariance spectrum with optional Marchenko-Pastur bounds (Marčenko and Pastur 1967) as an isotropic, white-noise reference for comparison. Multivariate normality is assessed through Mardia’s skewness and kurtosis tests (Mardia 1970), the Henze-Zirkler test (Henze and Zirkler 1990), and Royston’s procedure (Royston 1983). These tests combine marginal normality with inter-variable correlation structure to provide complementary assessments of multivariate Gaussianity; because they rely on a non-singular sample scatter matrix, they apply in the classical regime $p < n$ and require regularization or dimension reduction when $p \geq n$. Matrix conditioning is evaluated via the spectral condition number, numerical rank, and entropy-based effective rank (Roy and Vetterli 2007). Warnings are issued for ill-conditioned, rank-deficient, or strongly collinear matrices, which can compromise test performance and lead to unreliable inference.

3. scikit-covtest

3.1. Testing Methods

scikit-covtest version 0.1.1 is available on PyPI (pip install scikit-covtest) and on GitHub at <https://github.com/bystrogenomics/scikit-covtest> under the MIT license. All examples in this paper were produced with this version.

All hypothesis tests in **scikit-covtest** follow a single, lightweight interface aligned with **SciPy** style. Each function accepts an $n \times p$ data matrix and optional arguments that specify the null structure, and returns a Python dict with two keys: **stat** and **p_value**. For example, `leδοit_wolf_identity(X)` computes the test statistic under an identity null and returns `{"stat": ..., "p_value": ...}`. Inputs may be NumPy arrays or pandas DataFrames, and outputs are plain Python types that serialize cleanly for reproducible workflows.

The shared return format enables seamless composition with familiar tooling. Results can be tabulated across parameter grids, plotted against sample size or dimension, and passed into multiple testing procedures using the same field names. Because every test returns **stat** and **p_value**, benchmarking and automated reporting can be written once and reused across the library. This design lowers cognitive overhead, supports transparent evaluation, and encourages side-by-side comparisons of classical and high-dimensional procedures under a single, consistent API.

We have evaluated the methods implemented in **scikit-covtest** using standardized simulation studies. A representative subset of these evaluations is presented in Section 4, while complete, reproducible notebooks are available in the GitHub repository. Table 1 summarizes the currently implemented procedures across the four hypothesis classes considered in this package: identity, sphericity, proportionality, and two-sample covariance equality. For each method, the table indicates whether it is applicable in high-dimensional settings and the type of alternative to which it is primarily sensitive. As additional methods are incorporated, the same benchmarking framework can be used to extend these comparisons.

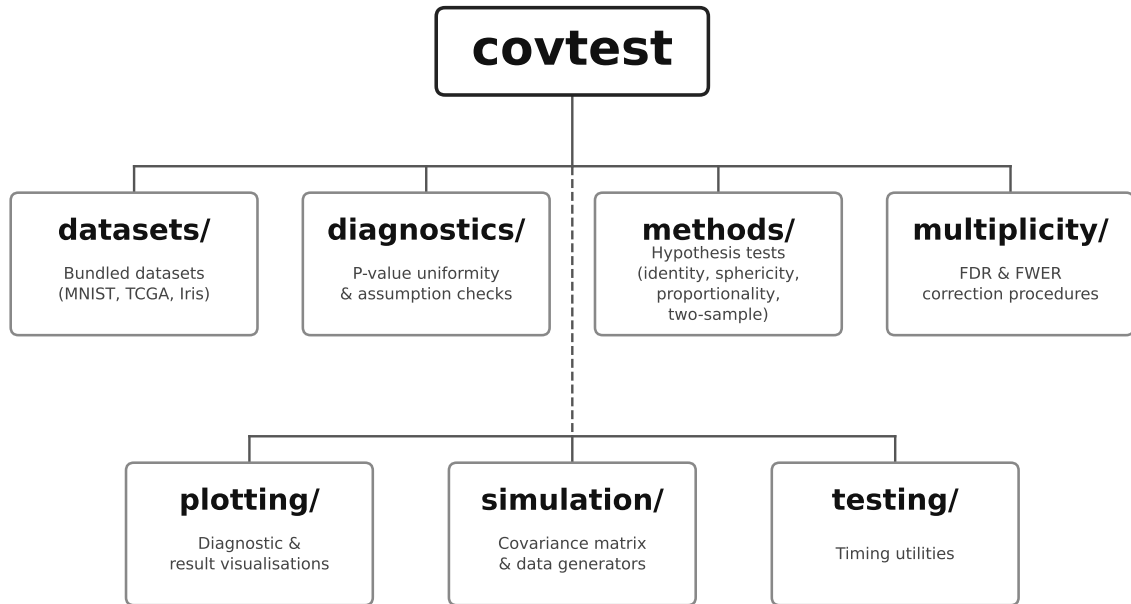


Figure 2: Directory structure of the **covtest** package. The root package contains seven sub-modules organised into two groups. The core group comprises **datasets** (bundled benchmark datasets: MNIST, TCGA, and Iris), **diagnostics** (p-value uniformity and distributional assumption checks), **methods** (implementations of hypothesis tests for identity, sphericity, proportionality, and two-sample covariance hypotheses), and **multiplicity** (FDR and FWER correction procedures). The support group comprises **plotting** (diagnostic and result visualisations), **simulation** (covariance matrix and data generators), and **testing** (internal timing utilities).

Table 1: Covariance structure tests by high-dimensional applicability and alternative sensitivity

Test	High Dimensional	Alternative Sensitivity
Identity Tests		
Nagao (1973)	No	Dense
Ledoit and Wolf (2002)	Yes	Dense
Srivastava (2005)	Yes	Dense
Chen, Zhang, and Zhong (2010)	Yes	Dense
Srivastava, Kollo, and von Rosen (2011)	Yes	Dense
Fisher (2012)	Yes	Dense
Srivastava, Yanagihara, and Kubokawa (2014)	Yes	Dense
Ahmad and von Rosen (2015)	Yes	Dense
Xu, Zhou, and Zhu (2025)	Yes	Dense
Sphericity Tests		
Bartlett (1937)	No	Dense
John (1971)	No	Dense
Srivastava (2005)	Yes	Dense
Muirhead (1982)	No	Dense
Fisher, Sun, and Gallagher (2010)	Yes	Low-rank/spiked
Chen <i>et al.</i> (2010)	Yes	Dense
Srivastava <i>et al.</i> (2014)	Yes	Dense
Ahmad and von Rosen (2015)	Yes	Dense
Xu <i>et al.</i> (2025)	Yes	Dense
Proportionality Tests		
Eriksen (1987)	No	Dense
Liu, Xu, Zheng, and Tian (2014)	Yes	Dense
Cheng, Liu, Peng, Zhang, and Zheng (2019)	Yes	Dense
Tsukuda and Matsuura (2019)	Yes	Dense
Ahmad (2022)	Yes	Dense
Two-Sample Tests		
Box (1949)	No	Dense
Schott (2001)	No	Dense
Schott (2007)	Yes	Dense
Srivastava (2007)	Yes	Dense
Srivastava <i>et al.</i> (2014)	Yes	Dense
Cai <i>et al.</i> (2013)	Yes	Sparse
Ishii (2017)	Yes	Low-rank/spiked
Chang, Zhou, Zhou, and Wang (2017)	Yes	Sparse
Ahmad (2017)	Yes	Dense
He and Chen (2018)	Yes	Sparse
Ding, Hu, and Wang (2025)	Yes	Low-rank/spiked
Li <i>et al.</i> (2025)	Yes	Dense

3.2. Multiple Hypothesis Testing

Multiple-hypothesis testing is a critical component of modern high-dimensional data analysis because it provides a framework for distinguishing reproducible signals from noise when

many hypotheses are tested simultaneously. This issue arises naturally when covariance tests are applied across many genes, feature blocks, experimental conditions, or population comparisons. Without multiplicity correction, the probability of reporting false-positive results increases rapidly with the number of tests.

The problem is especially prominent in omics applications, where researchers routinely perform thousands or millions of association tests. In genome-wide association studies (GWAS), for example, millions of single nucleotide polymorphisms may be evaluated for association with a phenotype. The conventional genome-wide significance threshold of $p < 5 \times 10^{-8}$ reflects the adaptation of family-wise error-rate control to this testing scale (Pe'er *et al.* 2008). More generally, appropriate control of the false discovery rate (FDR) or family-wise error rate (FWER) helps ensure that high-throughput studies produce reliable and reproducible findings (Dudbridge and Gusnanto 2008; Storey and Tibshirani 2003).

False Discovery Rate

The FDR is the expected proportion of false rejections among all rejected hypotheses. The Benjamini–Hochberg procedure (Benjamini and Hochberg 1995) controls the FDR under independence and certain positive-dependence conditions. The Benjamini–Yekutieli procedure (Benjamini and Yekutieli 2001) introduces a harmonic correction that provides control under arbitrary dependence, although the additional correction generally reduces power.

The module also implements procedures for incorporating external information or estimating the proportion of true null hypotheses. The weighted Benjamini–Hochberg procedure (Genovese *et al.* 2006) assigns nonnegative weights to the hypotheses, allowing hypotheses with stronger prior support to receive more favorable rejection thresholds. The implementation normalizes the weights internally so that they sum to the number of hypotheses.

The Storey–Tibshirani procedure (Storey and Tibshirani 2003) estimates the proportion π_0 of true null hypotheses from the upper tail of the observed p -value distribution. It evaluates the estimate over a user-specified or default grid of tuning parameters λ and uses the resulting estimate to calculate adjusted values. When $\hat{\pi}_0 < 1$, this can increase power relative to procedures that implicitly assume that all hypotheses are null.

Additional implemented methods include the step-down procedure of Benjamini and Liu (1999) and the weighted step-up construction of Blanchard and Roquain (2008). The latter accepts optional prior weights through the argument `pii`. These weights influence the rejection thresholds but do not constitute an estimate of π_0 .

Family-Wise Error Rate

The FWER is the probability of making at least one false rejection within a family of hypotheses. The Bonferroni and Holm procedures provide strong FWER control without requiring independence among the tests. Hochberg and Hommel procedures can provide greater power under stronger assumptions on the dependence structure.

The Romano–Wolf procedure (Romano and Wolf 2005) differs from the other implemented methods because it explicitly uses the joint null distribution of the test statistics. Rather than accepting marginal p -values, it takes observed test statistics and a matrix of bootstrap or permutation statistics generated under the joint null. This resampling approach can account for dependence among the hypotheses, provided that the resampling scheme adequately

represents their joint null distribution.

Table 2 summarizes the implemented procedures, whether they estimate π_0 , and their principal assumptions. A general review of multiple-testing methodology is provided by [Austin, Dialsingh, and Altman \(2014\)](#).

Table 2: Multiple-testing procedures implemented in **scikit-covtest**.

Procedure	Estimates π_0	Main requirement
<i>Family-wise error rate procedures</i>		
Dunn (1961)	No	Arbitrary dependence
Holm (1979)	No	Arbitrary dependence
Hochberg (1988)	No	Independence or positive dependence
Hommel (1988)	No	Independence or positive dependence
Romano and Wolf (2005)	No	Valid joint null resamples
<i>False discovery rate procedures</i>		
Benjamini and Hochberg (1995)	No	Independence or PRDS
Benjamini and Yekutieli (2001)	No	Arbitrary dependence
Genovese et al. (2006)	No	Fixed weights; independence or PRDS
Storey and Tibshirani (2003)	Yes	Independence
Benjamini and Liu (1999)	No	Independence
Blanchard and Roquain (2008)	No	Positive prior weights; arbitrary dependence

The procedures are divided between the **fdr** and **fwcr** submodules of **covtest.multiplicity**. The FDR procedures generally accept a vector of p -values and the desired error level **alpha**. Weighted Benjamini–Hochberg additionally accepts **weights**, the Blanchard–Roquain procedure accepts optional prior weights through **pii**, and the Storey–Tibshirani procedure accepts an optional grid of tuning parameters through **lambdas**.

The Bonferroni, Holm, Hochberg, and Hommel FWER procedures similarly accept marginal p -values and **alpha**. Romano–Wolf instead accepts the observed statistics **T_obs**, a matrix of jointly resampled statistics **T_boot**, and an optional **side** argument specifying the direction of the alternative.

All procedures return a dictionary containing **rejected**, **alpha**, and **method**. The FDR procedures return adjusted values under the key **qvals**, whereas the FWER procedures return adjusted p -values under **pvals_adj**. The Storey–Tibshirani procedure additionally returns the estimated null proportion under **pi0**. The Boolean vector **rejected** indicates which null hypotheses are rejected at the specified level; a value of **False** indicates failure to reject rather than acceptance of the null hypothesis.

```
>>> import numpy as np
>>> from covtest.multiplicity import fdr, fwer
>>> pvals = np.array([
...     0.001, 0.01, 0.03, 0.20, 0.35,
...     0.45, 0.65, 0.75, 0.85, 0.95,
... ])
>>> weights = np.array([
...     2.0, 2.0, 1.0, 1.0, 1.0,
...     1.0, 1.0, 1.0, 1.0, 1.0,
... ])
```

```

>>> res_bh = fdr.benjamini_hochberg(pvals, alpha=0.05)
>>> res_by = fdr.benjamini_yekutieli(pvals, alpha=0.05)
>>> res_wbh = fdr.weighted_bh(pvals, weights, alpha=0.05)
>>> res_storey = fdr.storey_qvalues(pvals, alpha=0.05)
>>> res_holm = fwer.holm(pvals, alpha=0.05)
>>> np.flatnonzero(res_bh["rejected"])
array([0, 1])
>>> np.flatnonzero(res_by["rejected"])
array([0])
>>> np.flatnonzero(res_wbh["rejected"])
array([0, 1])
>>> np.flatnonzero(res_storey["rejected"])
array([0, 1])
>>> round(float(res_storey["pi0"]), 3)
0.737
>>> np.round(res_holm["pvals_adj"][:3], 3)
array([0.01, 0.09, 0.24])

```

3.3. Synthetic Data Generation

Covariance Matrix Generation

We provide functions for generating structured covariance matrices, together with a Gaussian Orthogonal Ensemble generator for random-matrix and spectral experiments. The function `sample_goe(p)` samples a symmetric random matrix from the Gaussian Orthogonal Ensemble, with off-diagonal entries having variance one and diagonal entries having variance two. Unlike the covariance generators, a GOE matrix is generally not positive definite and is intended for random-matrix and spectral experiments rather than direct use as a covariance matrix. The function `sample_cov_uniform_correlation(K, level)` creates unit-variance correlation matrices with off-diagonal entries drawn uniformly from a range determined by `level`. The function `generate_spectral_cov(p)` constructs covariance matrices with randomly sampled eigenvalues and orthogonal bases, and `generate_toeplitz_cov(p, rho)` creates AR(1)-structured Toeplitz matrices with exponentially decaying correlations. For modular dependencies, `generate_block_diagonal_cov(p, block_size)` assembles block-diagonal matrices with positive semi-definite blocks, and `generate_low_rank_cov(p, rank, noise_var)` combines low-rank structure with isotropic Gaussian noise. To simulate sparse dependencies, `generate_sparse_precision_cov(p, sparsity)` samples a sparse precision matrix and returns its inverse covariance. The function `generate_marchenko_pastur(p, n)` samples from the white Wishart ensemble to approximate the Marcenko–Pastur distribution, while `generate_spiked_covariance(p, spike_eigenvalue, num_spikes)` creates spiked covariance matrices with a low-rank perturbation in the leading eigenspace. Together, these generators provide a flexible testbed for evaluating statistical methods under a range of spectral and structural covariance regimes.

```

>>> import numpy as np
>>> rng = np.random.default_rng(42)

```

```
>>> from covtest.simulation import generate_covariances as gc
>>> method = gc.generate_spiked_covariance
>>> options = dict(spike_eigenvalue=8.0, num_spikes=2)
>>> Sigma = method(p=5, rng=rng, **options)
```

Data Generation

We provide flexible generators for simulating multivariate heavy-tailed data under both null and alternative covariance structures, potentially with covariance matrices generated as above. The `generate_heavy_tailed_samples` function produces samples with exact covariance matching a user-supplied matrix, supporting a wide range of heavy-tailed distributions including Student's *t*, Laplace, log-normal, Pareto, and scale mixtures. For hypothesis testing or robustness evaluation, `generate_heavy_tailed_alternative` introduces structured deviations from the null through mean-shift mixtures, scaled covariance inflation, rank-one perturbations, or spiked eigenvalue bumps, while preserving heavy-tailed behavior. Both functions accept a shared interface with user-defined distribution parameters via options. As an example,

```
>>> import numpy as np
>>> rng = np.random.default_rng(42)
>>> from covtest.simulation import generate_data as gd
>>> from covtest.simulation import generate_covariances as gc
>>> Sigma = gc.generate_spectral_cov(p=5, rng=rng)
>>> method = gd.generate_heavy_tailed_samples
>>> options = dict(df=5)
>>> X = method(Sigma, n=100, dist_type="t", rng=rng, options=options)
>>> X = method(Sigma, n=100, dist_type="laplace", rng=rng)
>>> X = method(Sigma, n=100, dist_type="pareto", rng=rng, options={"b": 2.5})
>>> method = gd.generate_heavy_tailed_alternative
>>> alt_opts = dict(
...     alt="mixture",
...     dist={"df": 5},
...     mu_norm=1.0,
...     p_mix=0.5
... )
>>> result = method(Sigma, n=100, dist_type="t", rng=rng, options=alt_opts)
>>> result["alt"], result["X"].shape, np.round(result["cov_used"], 2)

('mixture', (100, 5), array([[..., ..., ...], ...]))
```

3.4. Test Diagnostics

The function `analyze_pvalues(pvals, num_permutations, seed)` computes a comprehensive suite of diagnostic statistics to assess whether a vector of *p*-values is consistent with a uniform distribution, as expected under a valid null. It includes classical tests such as the Kolmogorov–Smirnov ([Smirnov 1948](#)) test and an Anderson–Darling-like statistic tailored to the

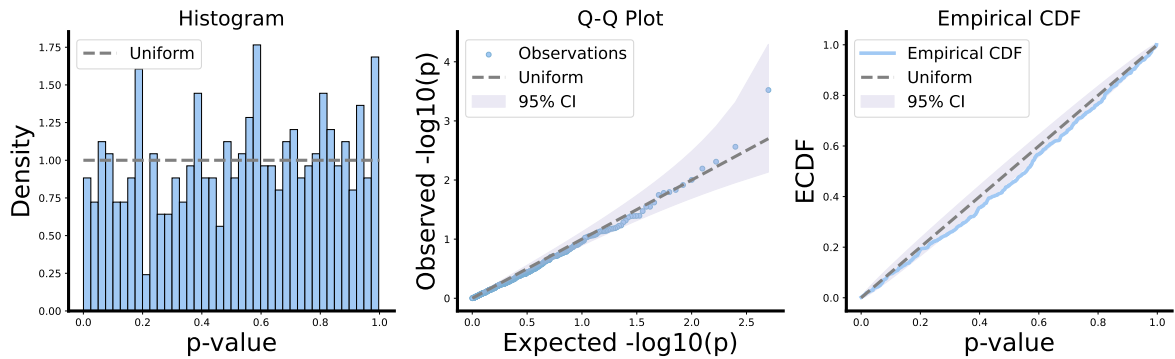


Figure 3: Diagnostic plots of the observed p -value distribution under the uniform null hypothesis. **Left:** Histogram of the observed p -values, with the uniform density shown by the dashed line. **Middle:** Q-Q plot of the observed versus theoretical $-\log_{10}(p)$ quantiles, with the pointwise 95% confidence band under the uniform null shown in gray. **Right:** Empirical cumulative distribution function (ECDF) of the observed p -values, together with the expected uniform CDF and its pointwise 95% confidence band.

Uniform[0,1] distribution, along with scalar deviation measures (L^∞ and L^2 norms) between the empirical and theoretical CDFs. Additional diagnostics include the genomic inflation factor (λ_{GC}), Storey's estimator of the null proportion (π_0), binomial enrichment tests for small- p tails, and a linear regression fit to the QQ plot (slope and intercept). A permutation-based p -value is also computed to assess the significance of the observed L^2 ECDF deviation. The function returns all results as a structured dictionary, enabling further tabulation or visualization.

```
>>> from covtest.diagnostics.evaluate_pvalues import analyze_pvalues
>>> import numpy as np
>>> pvals_null = np.random.default_rng(0).uniform(0, 1, size=500)
>>> result = analyze_pvalues(pvals_null, num_permutations=1000, seed=123)
>>> result.keys()
```

```
dict_keys(['ks', 'ad', 'ecdf_deviation', 'inflation_factor',
          'storey_pi0', 'tail_tests', 'qq_fit', 'perm_l2_pval'])
```

```
>>> from covtest.plotting.null import plot_pvalue_diagnostics_grid
>>> plot_pvalue_diagnostics_grid(pvals_null, sname='Figure3.pdf')
```

An example of the output is given in Figure 3. Each of the plots can be reproduced individually as well.

3.5. Data Loading

The functions `load_iris()`, `load_mnist()`, and `load_tcga()` provide convenient access to benchmark datasets stored as compressed NumPy archives. Only the small Iris archive is bundled inside the wheel; the two large archives, `mnist.npz` and `tcga.npz`, are hosted externally

(Zenodo, DOI 10.5281/zenodo.21600332) and are downloaded on first use via **pooch**, then cached in the user’s operating-system cache directory so that subsequent calls are immediate. Because **pooch** is an optional dependency, the remote datasets require installing the **datasets** extra:

```
pip install scikit-covtest[datasets]
```

Calling `load_mnist()` or `load_tcga()` without **pooch** installed raises an `ImportError` with installation instructions.

The function `load_mnist(split="train", return_X_y=True, normalize=True)` loads the MNIST handwritten digits dataset, returning either the training or the test split as a pair of arrays, or, when `return_X_y=False`, a dictionary containing both splits. When `normalize=True`, pixel values are cast to single precision and scaled to the range $[0, 1]$, and images are flattened to vectors of length 784 if they are stored as 28×28 arrays.

The function `load_tcga(return_X_y=True, return_names=False)` loads the gene-expression dataset described in Section 5.2. By default it returns the feature matrix and the vector of cancer-type labels. Setting `return_names=True` additionally returns the feature names, the name of the label column, and the sample identifiers; setting `return_X_y=False` returns the same five arrays as a dictionary.

```
>>> from covtest.datasets import load_mnist, load_tcga
>>> X, y = load_mnist(split="train", normalize=True)
>>> X, y, gene_names, label_names, sample_ids = load_tcga(return_names=True)
```

As distributed, the first column of the array returned by `load_tcga()` is the sample identifier rather than a gene, so the returned matrix has 20,532 columns of which 20,531 are gene-expression features. This column must be removed before numerical analysis; Section 5.2 shows the required step.

4. Null Calibration and Empirical Power

We evaluate a representative subset of the implemented tests across all four hypothesis classes using controlled Gaussian simulations. For each class we assess two properties: (i) size control under the null hypothesis, examined via probability–probability (P–P) plots of empirical against theoretical uniform p -values; and (ii) empirical power against a fixed alternative as the sample size increases. All data are generated using the **scikit-covtest** simulation utilities `generate_toeplitz_cov` and `generate_spiked_covariance`. The nominal level is $\alpha = 0.05$ throughout, with a fixed dimension $p = 20$ and Gaussian data. Simulations were generated with **scikit-covtest** version 0.1.1 using random seed 42; for replication r , the generator was initialized as `np.random.default_rng(42 + r)`. For the proportionality and two-sample settings, n denotes the sample size in each group.

4.1. Experimental design

Null calibration is assessed at $n = 200$ over 500 independent replications. Power is estimated over 300 replications at sample sizes $n \in \{50, 75, 100, 150, 200, 300, 400\}$. The code used to

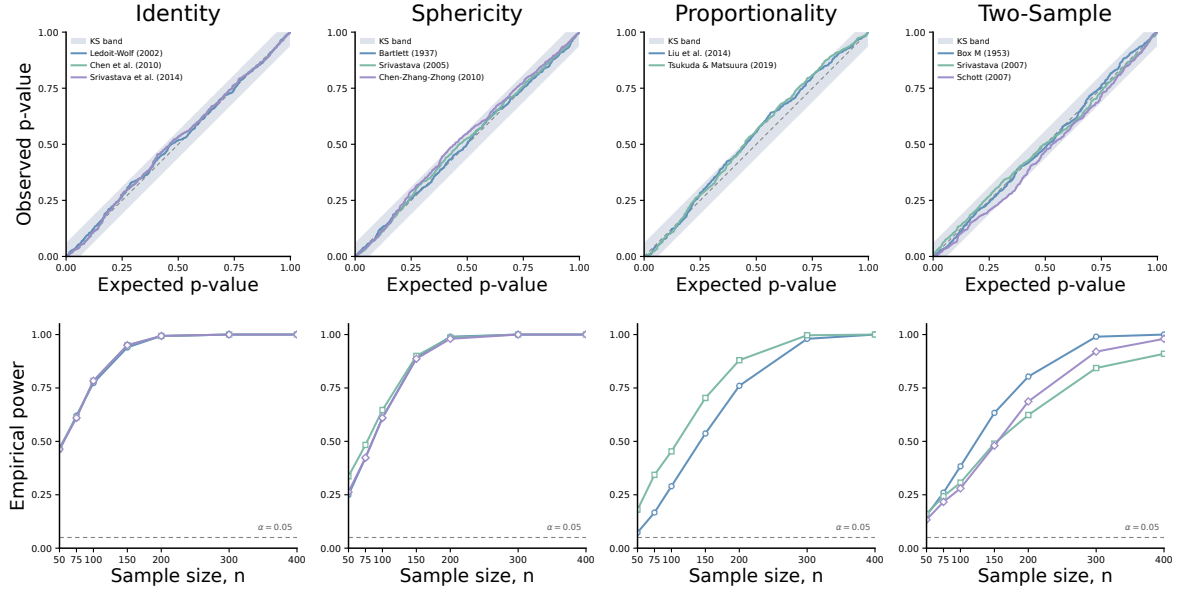


Figure 4: Null calibration (top row) and empirical power (bottom row) for a representative subset of tests across the four hypothesis classes in **scikit-covtest**, evaluated under Gaussian data with $p = 20$ dimensions. Top row: P-P plots of observed against expected p -values under the null. The shaded region is a 95% simultaneous Kolmogorov–Smirnov confidence band; the dashed line is the $y = x$ reference. Bottom row: empirical power at $\alpha = 0.05$.

generate Figure 4 is distributed as `figures/run_figure_simulations.py` in the versioned software archive. The null and alternative distributions for each class are as follows.

- Identity. Null: $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$. Alternative: $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\text{spike}})$, a spiked model with two leading eigenvalues equal to 1.8 and the remaining $p - 2$ eigenvalues equal to 1.0.
- Sphericity. Null: $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, 3\mathbf{I}_p)$. Alternative: $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma_T(0.15))$, an AR(1) Toeplitz matrix with off-diagonal decay parameter $\rho = 0.15$, which is not proportional to the identity.
- Proportionality. Null: $(\mathbf{x}, \mathbf{y})$ drawn from $\mathcal{N}(\mathbf{0}, \Sigma_T(0.3))$ and $\mathcal{N}(\mathbf{0}, 2\Sigma_T(0.3))$ respectively, satisfying $\Sigma_1 = c\Sigma_2$ with $c = 1/2$. Alternative: $\mathbf{y}$ is redrawn from $\mathcal{N}(\mathbf{0}, \Sigma_T(0.12))$, which is not proportional to $\Sigma_T(0.3)$ because the lag- k entry ratio $(0.3/0.12)^k$ varies with k .
- Two-sample. Null: $\Sigma_1 = \Sigma_2 = \Sigma_T(0.3)$. Alternative: $\Sigma_2 = \Sigma_T(0.45)$, a denser correlation structure.

4.2. Null calibration

Results are shown in the top row of Figure 4. For the identity tests, all three methods, Ledoit and Wolf (2002), Chen *et al.* (2010), and Srivastava *et al.* (2014), track the diagonal closely

and remain within the 95% simultaneous Kolmogorov–Smirnov confidence band, indicating well-controlled Type I error at this dimension and sample size. The three sphericity tests [Bartlett \(1937\)](#), [Srivastava \(2005\)](#), and [Chen *et al.* \(2010\)](#) are likewise close to the diagonal, with only a marginal excursion of the [Chen *et al.* \(2010\)](#) curve at the upper end of the band.

Among the proportionality tests, both methods are conservative at this combination of n and p : observed p -values systematically exceed their expected uniform quantiles, so both reject less often than the nominal level warrants. The departure is somewhat more pronounced for [Tsukuda and Matsuura \(2019\)](#) than for [Liu *et al.* \(2014\)](#), though neither procedure tracks the diagonal as tightly as the identity or sphericity tests in this regime.

For the two-sample class, [Box \(1949\)](#) is the best calibrated of the three procedures examined, tracking the diagonal closely across the full range with no notable excursion outside the confidence band. [Schott \(2007\)](#) is the only method whose curve leaves the band, drifting mildly anti-conservative, consistent with its reliance on a standard normal limit that is not yet fully accurate at this combination of dimension and sample size. [Srivastava \(2007\)](#) trends in the opposite direction, staying within the band but rejecting somewhat less often than the nominal rate would suggest.

4.3. Empirical power

Power curves are shown in the bottom row of Figure 4. For the identity class, the two high-dimensional U-statistic tests of [Chen *et al.* \(2010\)](#) and [Srivastava *et al.* \(2014\)](#) achieve moderately higher power than Ledoit–Wolf (2002) at small n , with all three converging to near-perfect detection by $n \approx 200$. With the corrected Ledoit–Wolf degrees-of-freedom implementation, its empirical power is 0.46, 0.62, and 0.773 at $n = 50, 75$, and 100, respectively. For sphericity, all three methods produce similar power curves; the mild Toeplitz alternative ($\rho = 0.15$) constitutes a low-magnitude departure from sphericity that requires $n \approx 300$ for reliable detection regardless of method.

For proportionality, [Tsukuda and Matsuura \(2019\)](#) attains higher power than [Liu *et al.* \(2014\)](#) across all sample sizes examined. We report this difference descriptively: both procedures are conservative under the null at this (n, p) , so the gap cannot be attributed to a calibration asymmetry between them. For the two-sample class, [Box \(1949\)](#) attains the highest power of the three procedures from $n = 100$ onward, reaching 0.803 at $n = 200$ and 1.000 at $n = 400$, compared with 0.623 and 0.910 for [Srivastava \(2007\)](#) and 0.687 and 0.980 for [Schott \(2007\)](#). Because Box’s M exhibits no evident size inflation in this design (Section 4.2), this advantage is consistent with greater sensitivity in this simulated setting.

5. Worked Example with Included Datasets

5.1. Size and Power with MNIST

We use the MNIST handwritten-digit dataset ([LeCun *et al.* 1998](#)) as a high-dimensional, non-Gaussian illustration of two-sample covariance testing. Each image is represented by $28 \times 28 = 784$ pixel intensities. We combine the training and test splits, scale pixel intensities to $[0, 1]$ using the dataset loader, and then standardize each pixel separately within the digit-0 and digit-1 samples. Values above 10 after standardization are truncated to 10. This

preprocessing focuses the comparison on differences in within-class correlation structure rather than marginal location and scale.

We compare the Srivastava (2007) and Srivastava–Yanagihara (2014) two-sample covariance tests. For null calibration, each of 300 replications draws $2N$ distinct digit-0 images without replacement and partitions them into two nonoverlapping samples of size $N = 200$. For the alternative, one sample of N digit-0 images and one independently drawn sample of N digit-1 images are compared. All replications use `np.random.default_rng(0)`. The complete analysis and figure-generation code are provided in `figures/run_mnist_example.py`.

```
>>> import numpy as np
>>> from sklearn.preprocessing import StandardScaler
>>> from covtest.datasets import load_mnist
>>> from covtest.methods.hypothesis_two_sample import (
...     srivastava_two_sample_2007,
...     srivastava_yanagihara_two_sample,
... )
>>> N, n_rep = 200, 300
>>> X_tr, y_tr = load_mnist(split="train")
>>> X_te, y_te = load_mnist(split="test")
>>> X = np.concatenate((X_tr, X_te))
>>> y = np.concatenate((y_tr, y_te))
>>> X0 = StandardScaler().fit_transform(X[y == 0])
>>> X1 = StandardScaler().fit_transform(X[y == 1])
>>> X0[X0 > 10] = 10
>>> X1[X1 > 10] = 10
>>> methods = [
...     srivastava_two_sample_2007,
...     srivastava_yanagihara_two_sample,
... ]
>>> rng = np.random.default_rng(0)
>>> null_pvalues = np.empty((n_rep, len(methods)))
>>> alternative_pvalues = np.empty_like(null_pvalues)
>>> for i in range(n_rep):
...     idx0 = rng.choice(X0.shape[0], size=2 * N, replace=False)
...     idx1 = rng.choice(X1.shape[0], size=N, replace=False)
...     for j, method in enumerate(methods):
...         null_pvalues[i, j] = method(X0[idx0[:N]], X0[idx0[N:]])["p_value"]
...         alternative_pvalues[i, j] = method(X0[idx0[:N]], X1[idx1])["p_value"]
```

Figure 5 summarizes the input distributions, null calibration, and empirical rejection rates at $\alpha = 0.05$. The pixel-intensity distributions are strongly non-Gaussian, with substantial mass near zero and a right-skewed positive tail. Under the digit-0 versus digit-0 null, the empirical rejection rates are 0.023 for Srivastava (2007) and 0.003 for Srivastava–Yanagihara (2014). Thus, both procedures are conservative in this non-Gaussian design, particularly the latter. Under the digit-0 versus digit-1 comparison, the corresponding empirical rejection rates are 0.687 and 1.000. These results illustrate that the two methods can detect marked differences in the covariance structure of the two digit classes, while also showing the importance of

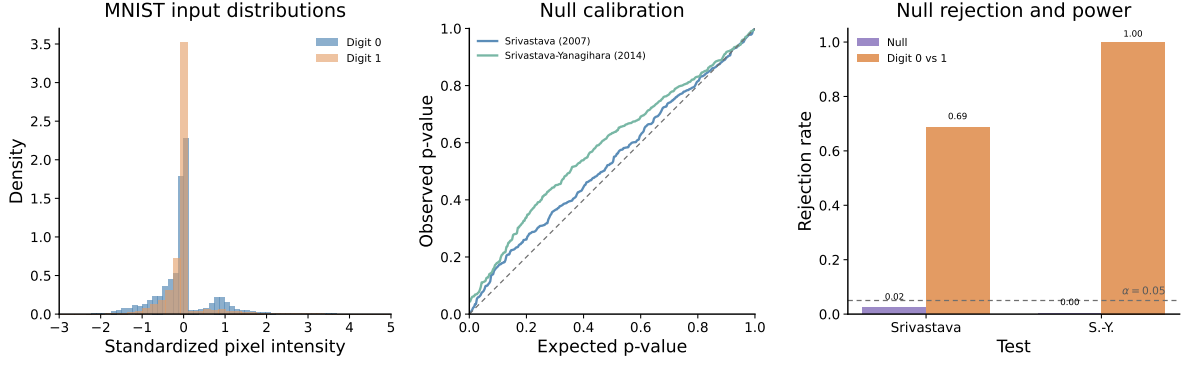


Figure 5: MNIST two-sample covariance-test results for digit 0 versus digit 1. The left panel shows the standardized pixel-intensity distributions. The middle panel presents P–P plots for the digit-0 versus digit-0 null comparison. The right panel reports empirical rejection rates at $\alpha = 0.05$ under the null and under the digit-0 versus digit-1 alternative, based on 300 replications with $N = 200$ observations per group.

checking null calibration when applying large-sample procedures to structured, non-Gaussian image data.

5.2. Multiple Hypothesis Testing with TCGA

In our second example, we demonstrate how **scikit-covtest** combines two-sample covariance-structure tests with multiple-testing procedures. We use the gene-expression cancer RNA-Seq dataset (Fiorini 2016), a random extraction from the RNA-Seq (HiSeq) PANCAN dataset assembled by The Cancer Genome Atlas Pan-Cancer Analysis Project (Weinstein, Collisson, Mills, Shaw, Ozenberger, Ellrott, Shmulevich, Sander, and Stuart 2013). The dataset contains 801 samples and 20,531 gene-expression features. We compare breast invasive carcinoma (BRCA, $n = 300$) with lung adenocarcinoma (LUAD, $n = 141$).

To target differences in correlation rather than marginal variances, we standardize every expression feature separately within BRCA and LUAD. We then partition the 20,531 features into 4,106 nonoverlapping blocks of five consecutive features, discarding the final trailing feature. The first column returned by `load_tcga()` is a sample identifier and is removed before the expression matrix is converted to numeric form.

The Srivastava (2007) statistic is undefined for blocks whose pooled trace-square covariance estimate is zero. Of the 4,106 initial blocks, 30 are degenerate in this sense and are excluded before multiplicity correction. The remaining 4,076 valid block-level p -values form the family used for the Bonferroni, Benjamini–Hochberg (BH), and Benjamini–Yekutieli (BY) adjustments. The complete analysis and figure-generation code are provided in `figures/run_tcga_example.py`.

```
>>> import numpy as np
>>> from sklearn.preprocessing import StandardScaler
>>> from covtest.datasets import load_tcga
>>> from covtest.methods.hypothesis_two_sample import srivastava_two_sample_2007
>>> from covtest.multiplicity.fdr import benjamini_hochberg, benjamini_yekutieli
```

```

>>> from covtest.multiplicity.fwer import bonferroni
>>> X_raw, y = load_tcga()
>>> X = X_raw[:, 1:].astype(float) # Drop the sample-ID column.
>>> X_brca = StandardScaler().fit_transform(X[y == "BRCA"])
>>> X_luad = StandardScaler().fit_transform(X[y == "LUAD"])
>>> n_groups = X.shape[1] // 5
>>> pvalues = np.full(n_groups, np.nan)
>>> for i in range(n_groups):
...     idx = 5 * i + np.arange(5)
...     try:
...         pvalues[i] = srivastava_two_sample_2007(
...             X_brca[:, idx], X_luad[:, idx]
...         )["p_value"]
...     except ValueError:
...         pass # Degenerate block; exclude before multiplicity correction.
>>> valid = np.isfinite(pvalues)
>>> tested_pvalues = pvalues[valid]
>>> bonf = bonferroni(tested_pvalues)
>>> res_bh = benjamini_hochberg(tested_pvalues)
>>> res_by = benjamini_yekutieli(tested_pvalues)

```

At $\alpha = 0.05$, Bonferroni rejects 23 of the 4,076 valid blocks. BH rejects 41 blocks, whereas the more conservative BY procedure rejects 26 blocks. The ordering $23 \leq 26 \leq 41$ is expected: Bonferroni controls the family-wise error rate, whereas BH and BY control the false discovery rate, with BY additionally accounting for arbitrary dependence through its harmonic correction. Figure 6 displays the block-level p -values and the BH and BY adjusted p -values. Values with $-\log_{10}$ adjusted p -values at or above 20 are displayed at 20 to avoid numerical underflow dominating the plot.

6. Conclusion

In this work, we introduce **scikit-covtest**, a comprehensive Python package for hypothesis testing of covariance matrices. Our primary objective is to provide reliable, well-tested implementations of a wide range of hypothesis tests drawn from the statistical literature. To support researchers and practitioners, we include clear demonstrations using synthetic data, systematic performance comparisons across diverse statistical regimes, and illustrative applications using real datasets.

The package is organized into components to promote clarity and extensibility. The core module, **methods**, implements the hypothesis tests. The **diagnostics** module provides tools for assessing the plausibility of test assumptions and identifying potential violations. The **multiplicity** module addresses multiple hypothesis testing, a critical consideration in high-dimensional and large-scale applications. The **simulation** module generates synthetic data under both null models and common alternatives, enabling controlled benchmarking. Finally, the **testing** module streamlines systematic evaluations and performance comparisons of selected methods.

Although we incorporated a broad range of available tests, some were undoubtedly omit-

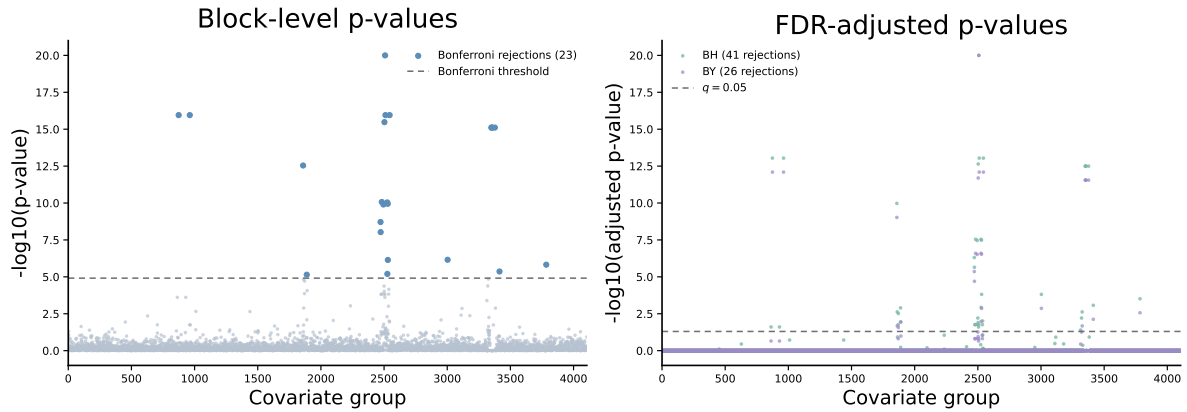


Figure 6: TCGA results for the BRCA-versus-LUAD comparison. The left panel shows block-level p -values for the 4,076 valid five-feature blocks; the dashed line denotes the Bonferroni threshold at $\alpha = 0.05$, and highlighted points are Bonferroni rejections. The right panel shows BH and BY adjusted p -values. Thirty degenerate blocks were excluded before multiplicity correction.

ted. Furthermore, the field is evolving and additional methods will inevitably be developed. We plan to maintain and expand **scikit-covtest** as new approaches are proposed, and we welcome contributions from the community through our public repository (<https://github.com/bystrogenomics/scikit-covtest>).

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